🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.

Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Convert Measurements Between Systems | Conversion factor | Customary system | Metric system | Approximately | Ratio table

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. Rewriting a customary measurement as a metric one gives an ________ answer, not an exact one.

2. The ratio that trades one unit for another is a ________ ________.

3. Inches, pounds and gallons belong to the ________ system.

4. Centimeters, grams and liters belong to the ________ system, built on tens.

5. Close to but not exactly equal is ________ — written with ≈.

6 A table whose every column holds the same comparison is a

______ ______.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Convert Measurements Between Systems** — Rewriting a customary measurement as a metric one, or the reverse, so two amounts can be compared.
  Convertir medidas entre sistemas — Reescribir una medida usual como métrica, o al revés, para poder comparar dos cantidades.

- [ ] **Conversion factor** — The ratio that links a unit in one system to a unit in the other, such as 1 mile to about 1.609 kilometers.
  Factor de conversión — La razón que relaciona una unidad de un sistema con una del otro, como 1 milla a unos 1.609 kilómetros.

- [ ] **Customary system** — The measurement system used in the United States: inches, feet, miles, ounces, pounds, cups, quarts, gallons.
  Sistema usual — El sistema de medidas que se usa en Estados Unidos: pulgadas, pies, millas, onzas, libras, tazas, cuartos y galones.

- [ ] **Metric system** — The measurement system used in most of the world, built on tens: centimeters, meters, kilometers, grams, kilograms, liters.
  Sistema métrico — El sistema de medidas que se usa en casi todo el mundo, basado en decenas: centímetros, metros, kilómetros, gramos, kilogramos y litros.

- [ ] **Approximately** — Close to, but not exactly equal. Conversions between systems are almost always approximate, so they use the ≈ sign.
  Aproximadamente — Cercano, pero no exactamente igual. Las conversiones entre sistemas casi siempre son aproximadas y usan el signo ≈.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The conversion factor is 1 km ≈ ____ mi. Scaling that up, 100 kilometers ≈ ____ miles, so the sign means about ____ miles per hour. That is ____ than 100 miles per hour, because in one hour you would travel ____ distance.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Divide the total cost by the number of pounds to find the cost per 1 pound.

Evidence: Students divide $18 by 3 to get $6 per pound and explain the unit rate lets them find the cost of any number of pounds.

Reasoning: This evidence connects the definition of convert measurements between systems to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The conversion factor is 1 km ≈ ____ mi. Scaling that up, 100 kilometers ≈ ____ miles, so the sign means about ____ miles per hour. That is ____ than 100 miles per hour, because in one hour you would travel ____ distance.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurements Between Systems
2. **Notes 2:** Conversion factor
3. **Notes 3:** Customary system
4. **Notes 4:** Metric system
5. **Notes 5:** Approximately
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake converting between systems is comparing the bare numbers without converting at all — deciding that 5 kilometers and 5 miles are the same distance because both say 5. They are not: 5 kilometers ≈ 3 miles. A second mistake is writing = instead of ≈. Conversion factors between systems are rounded (1 mile is about 1.609344 kilometers), so the result is approximate no matter how carefully you multiply.*
8. **Try It 1:** — *Each pair is a conversion factor between the two systems, and each one is approximate.*
9. **Try It 2:** A. The conversion factors are rounded, so results are approximate — *One mile is about 1.609344 kilometers — a decimal that never ends neatly. Any usable factor is rounded, so the answer is approximate.*